CEE6501 — Lecture 13.1

Force Density Method (Linear Examples)

Learning Objectives

By the end of this lecture, you will be able to:

- explain

Agenda

- Part 1 -

Part 1 - Review of Force Density (Linear)

Define

$$\mathbf{D} = \mathbf{C}^T\mathbf{Q}\mathbf{C}, \qquad \mathbf{D}_f = \mathbf{C}^T\mathbf{Q}\mathbf{C}_f$$

Doing this for all 3 directions, the force density equations become

$$\mathbf{D}\mathbf{x} = \mathbf{p}_x - \mathbf{D}_f\mathbf{x}_f \tag{4-1}$$

$$\mathbf{D}\mathbf{y} = \mathbf{p}_y - \mathbf{D}_f\mathbf{y}_f \tag{4-2}$$

$$\mathbf{D}\mathbf{z} = \mathbf{p}_z - \mathbf{D}_f\mathbf{z}_f \tag{4-3}$$

These are **linear equations in the unknown coordinates**.

The Key Advantage: A Linear Formulation

For prescribed:

- connectivity
- support coordinates
- nodal loads
- force densities

the unknown free-node coordinates follow from a **single linear solve**.

That was the original appeal of the force density method.

In a planar case, once we solve

$$\mathbf{x} = \mathbf{D}^{-1}\left(\mathbf{p}_x - \mathbf{D}_f\mathbf{x}_f\right) \tag{5-1}$$

$$\mathbf{y} = \mathbf{D}^{-1}\left(\mathbf{p}_y - \mathbf{D}_f\mathbf{y}_f\right) \tag{5-2}$$

and for 3D we add in

$$\mathbf{z} = \mathbf{D}^{-1}\left(\mathbf{p}_z - \mathbf{D}_f\mathbf{z}_f\right) \tag{5-3}$$

Once $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$ are solved, the new equilibrium geometry is fully determined.

From there, we can:

1. reconstruct the updated nodal geometry
2. compute the corresponding branch lengths
3. recover the branch forces from

$$\mathbf{s} = \mathbf{L}\mathbf{q}$$

This is why the force density method is often described as a **one-step method**: once the force densities $\mathbf{q}$ are prescribed, the equilibrium geometry follows from a linear solve.

Solution Procedure

For a simple planar example with two free nodes, the workflow is:

1. choose the connectivity and build $\mathbf{C}$ and $\mathbf{C}_f$
2. prescribe the force densities and build $\mathbf{Q}$ (user chooses)
3. compute

$$\mathbf{D} = \mathbf{C}^T \mathbf{Q} \mathbf{C}$$

   and

$$\mathbf{D}_f = \mathbf{C}^T \mathbf{Q} \mathbf{C}_f$$

4. solve separately for the unknown x- and y-coordinates
5. recover branch lengths and then branch forces

Repository Link

https://github.com/Bruun-Automation-Research-Lab/matrix-form-finding

Running Code

Run from the `./formfinder/main.py` file

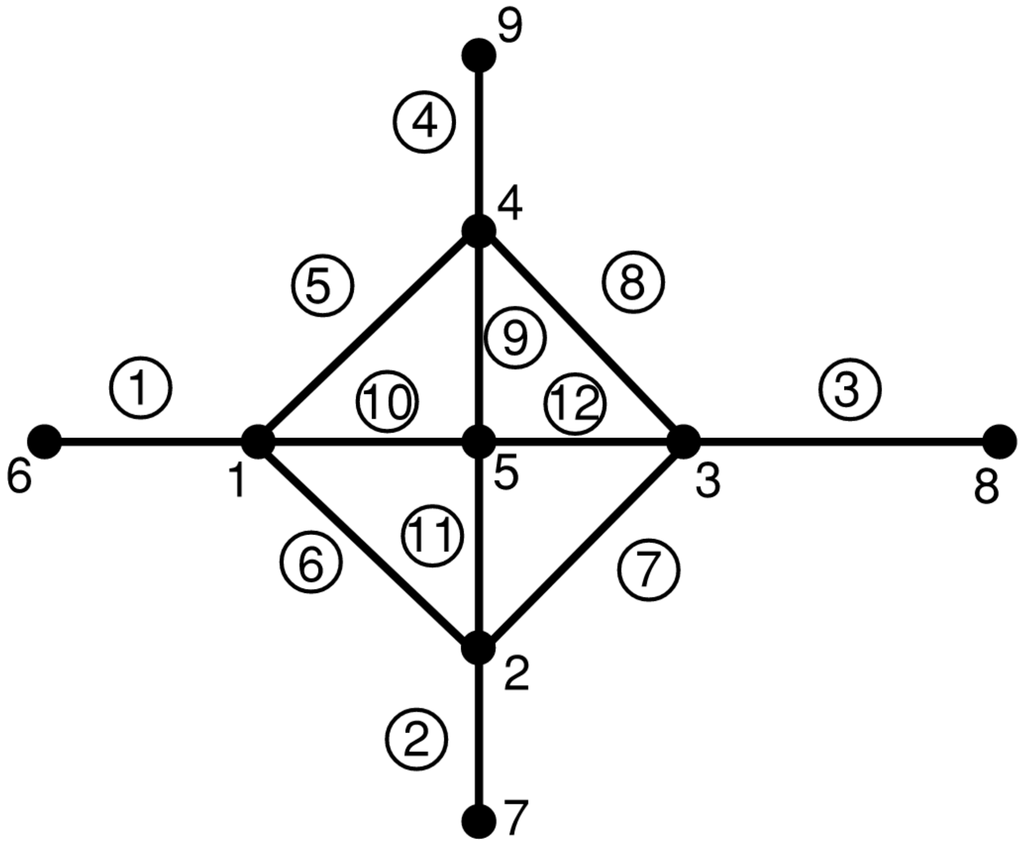
Input files specified in `./formfinder/structures` folder

Text data saved to `./debug` folder

Output plots and daya saved to Text data saved to `./output` folder

Part 2 - Simple 2D Example (Schek Paper)

Description of Structure



Geometry Setup

NODES

[n x 3]

	x	y	z
	1 x	1 x	1 x
1	-2.00	0.00	0.00
2	0.00	-2.00	0.00
3	2.00	0.00	0.00
4	0.00	2.00	0.00
5	0.00	0.00	0.00
6	-3.00	0.00	0.00
7	0.00	-3.00	0.00
8	3.00	0.00	0.00
9	0.00	3.00	0.00

ELEMENTS

[m x 2] & [m x 1]

	n_i	n_j
	1 x	1 x
1	1	6
2	2	7
3	3	8
4	4	9
5	1	4
6	1	2
7	2	3
8	3	4
9	4	5
10	1	5
11	2	5
12	3	5

NODAL LOADS

[n x 3]

	p_x	p_y	p_z
	1 x	1 x	1 x
1	0.00	0.00	0.00
2	0.00	0.00	0.00
3	0.00	0.00	0.00
4	0.00	0.00	0.00
5	0.00	0.00	0.00
6	0.00	0.00	0.00
7	0.00	0.00	0.00
8	0.00	0.00	0.00
9	0.00	0.00	0.00

NODAL FIXITY

[n x 1]

1 = fixed | 0 = free

	fix?
	1 x
1	0
2	0
3	0
4	0
5	0
6	1
7	1
8	1
9	1

Geometry Setup (Branch-Node Matrices)

```
BRANCH-NODE MATRIX
[m x n]
C_total =
| | |   1   2   3   4   5   6   7   8   9
----+------------------------------------
| | |   1   2   3   4   5   6   7   8   9
----+------------------------------------
  1 |   1   .   .   .   .  -1   .   .   .
  2 |   .   1   .   .   .   .  -1   .   .
  3 |   .   .   1   .   .   .   .  -1   .
  4 |   .   .   .   1   .   .   .   .  -1
  5 |   1   .   .  -1   .   .   .   .   .
  6 |   1  -1   .   .   .   .   .   .   .
  7 |   .   1  -1   .   .   .   .   .   .
  8 |   .   .   1  -1   .   .   .   .   .
  9 |   .   .   .   1  -1   .   .   .   .
 10 |   1   .   .   .  -1   .   .   .   .
 11 |   .   1   .   .  -1   .   .   .   .
 12 |   .   .   1   .  -1   .   .   .   .
```

```
BRANCH-NODE MATRIX (FREE)
[m x n_free]
C =
| | |   1   2   3   4   5
----+--------------------
| | |   1   2   3   4   5
----+--------------------
  1 |   1   .   .   .   .
  2 |   .   1   .   .   .
  3 |   .   .   1   .   .
  4 |   .   .   .   1   .
  5 |   1   .   .  -1   .
  6 |   1  -1   .   .   .
  7 |   .   1  -1   .   .
  8 |   .   .   1  -1   .
  9 |   .   .   .   1  -1
 10 |   1   .   .   .  -1
 11 |   .   1   .   .  -1
 12 |   .   .   1   .  -1
```

```
BRANCH-NODE MATRIX (FIXEI
[m x n_fixed]
C_f =
| | |   1   2   3   4
----+----------------
| | |   6   7   8   9
----+----------------
  1 |  -1   .   .   .
  2 |   .  -1   .   .
  3 |   .   .  -1   .
  4 |   .   .   .  -1
  5 |   .   .   .   .
  6 |   .   .   .   .
  7 |   .   .   .   .
  8 |   .   .   .   .
  9 |   .   .   .   .
 10 |   .   .   .   .
 11 |   .   .   .   .
 12 |   .   .   .   .
```

Option 1 — Uniform Force Density

Assume the same force density for every member: $q_i = 1$

So the force density vector becomes

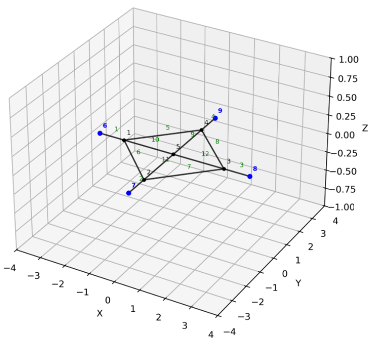
$$\mathbf{q} = [1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1]^T$$

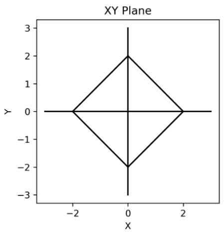
Using this, we form the matrices

$$\mathbf{D} = \mathbf{C}^T\mathbf{Q}\mathbf{C} \qquad \mathbf{D}_f = \mathbf{C}^T\mathbf{Q}\mathbf{C}_f$$

```
D (FREE)
[n_free x n_free]
D =
1 x
 |   |     |      1      2      3      4      5
 |   |     |      1      2      3      4      5
 ----+-----+-----------------------------------
 | 1 |   1 |   4.00  -1.00      .  -1.00  -1.00
 | 2 |   2 |  -1.00   4.00  -1.00      .  -1.00
 | 3 |   3 |      .  -1.00   4.00  -1.00  -1.00
 | 4 |   4 |  -1.00      .  -1.00   4.00  -1.00
 | 5 |   5 |  -1.00  -1.00  -1.00  -1.00   4.00
```

```
D (FIXED)
[n_free x n_fixed]
D_f =
1 x
 |   |     |      1      2      3      4
 |   |     |      6      7      8      9
 ----+-----+----------------------------
 | 1 |   1 |  -1.00      .      .      .
 | 2 |   2 |      .  -1.00      .      .
 | 3 |   3 |      .      .  -1.00      .
 | 4 |   4 |      .      .      .  -1.00
 | 5 |   5 |      .      .      .      .
```



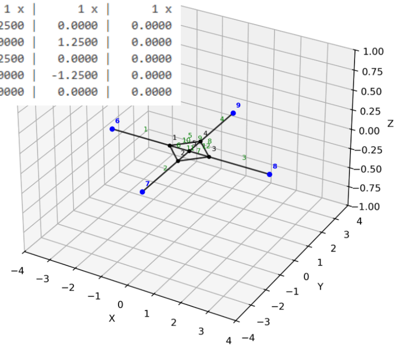


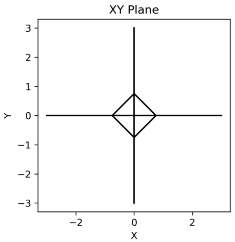
NODES
[n x 3]

	x 1 x	y 1 x	z 1 x
1	-2.00	0.00	0.00
2	0.00	-2.00	0.00
3	2.00	0.00	0.00
4	0.00	2.00	0.00
5	0.00	0.00	0.00
6	-3.00	0.00	0.00
7	0.00	-3.00	0.00
8	3.00	0.00	0.00
9	0.00	3.00	0.00

DELTA DISPLACEMENTS (FREE)
[n_free x 1]

	Node	d_x 1 x	d_y 1 x	d_z 1 x
1	1	1.2500	0.0000	0.0000
2	2	0.0000	1.2500	0.0000
3	3	-1.2500	0.0000	0.0000
4	4	0.0000	-1.2500	0.0000
5	5	0.0000	0.0000	0.0000





NODES (NEW)
[n x 3]

	x 1 x	y 1 x	z 1 x
1	-0.75	0.00	0.00
2	0.00	-0.75	0.00
3	0.75	0.00	0.00
4	0.00	0.75	0.00
5	0.00	0.00	0.00
6	-3.00	0.00	0.00
7	0.00	-3.00	0.00
8	3.00	0.00	0.00
9	0.00	3.00	0.00

FINAL NODES

[n x 3]

	x	y	z
	1 x	1 x	1 x
1	-0.750	-0.000	0.000
2	-0.000	-0.750	0.000
3	0.750	-0.000	0.000
4	-0.000	0.750	0.000
5	-0.000	-0.000	0.000
6	-3.000	0.000	0.000
7	0.000	-3.000	0.000
8	3.000	0.000	0.000
9	0.000	3.000	0.000

FINAL FORCE/LENGTH/FORCE DENSITY

[m x 1]

	Force	Length	q
	1 x	1 x	1 x
1	2.2500	2.2500	1.0000
2	2.2500	2.2500	1.0000
3	2.2500	2.2500	1.0000
4	2.2500	2.2500	1.0000
5	1.0607	1.0607	1.0000
6	1.0607	1.0607	1.0000
7	1.0607	1.0607	1.0000
8	1.0607	1.0607	1.0000
9	0.7500	0.7500	1.0000
10	0.7500	0.7500	1.0000
11	0.7500	0.7500	1.0000
12	0.7500	0.7500	1.0000

Option 2 — Scaled Force Density

Scale all force densities for all members uniformly: $q_i = 5$

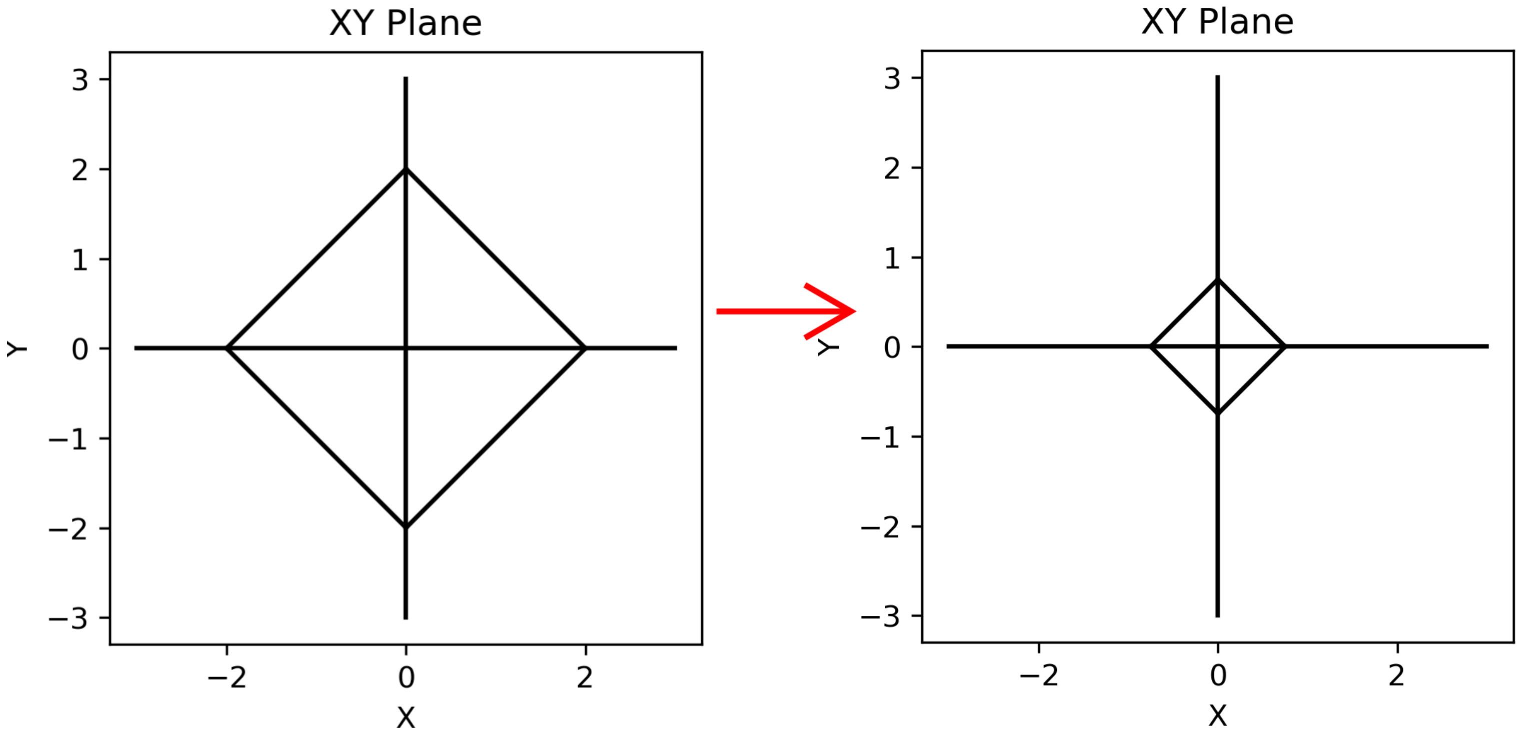
So the force density vector becomes

$$\mathbf{q} = [5,\ 5,\ 5,\ 5,\ 5,\ 5,\ 5,\ 5,\ 5,\ 5,\ 5,\ 5]^T$$

This gives the same *form* as Option 1.

Key idea:
Only the *relative* values of q matter — not their absolute magnitude.

Scaling all q by a constant changes the force level, but not the geometry.


XY Plane
Y
X
3
2
1
0
−1
−2
−3
−2
0
2
XY Plane
Y
X

Final nodal position the same as in the $q = 1$ case

FINAL NODES
[n x 3]

	x	y	z
	1 x	1 x	1 x
1	-0.750	-0.000	0.000
2	-0.000	-0.750	0.000
3	0.750	-0.000	0.000
4	-0.000	0.750	0.000
5	-0.000	-0.000	0.000
6	-3.000	0.000	0.000
7	0.000	-3.000	0.000
8	3.000	0.000	0.000
9	0.000	3.000	0.000

FINAL FORCE/LENGTH/FORCE DENSITY
[m x 1]

	Force	Length	q
	1 x	1 x	1 x
1	11.2500	2.2500	5.0000
2	11.2500	2.2500	5.0000
3	11.2500	2.2500	5.0000
4	11.2500	2.2500	5.0000
5	5.3033	1.0607	5.0000
6	5.3033	1.0607	5.0000
7	5.3033	1.0607	5.0000
8	5.3033	1.0607	5.0000
9	3.7500	0.7500	5.0000
10	3.7500	0.7500	5.0000
11	3.7500	0.7500	5.0000
12	3.7500	0.7500	5.0000

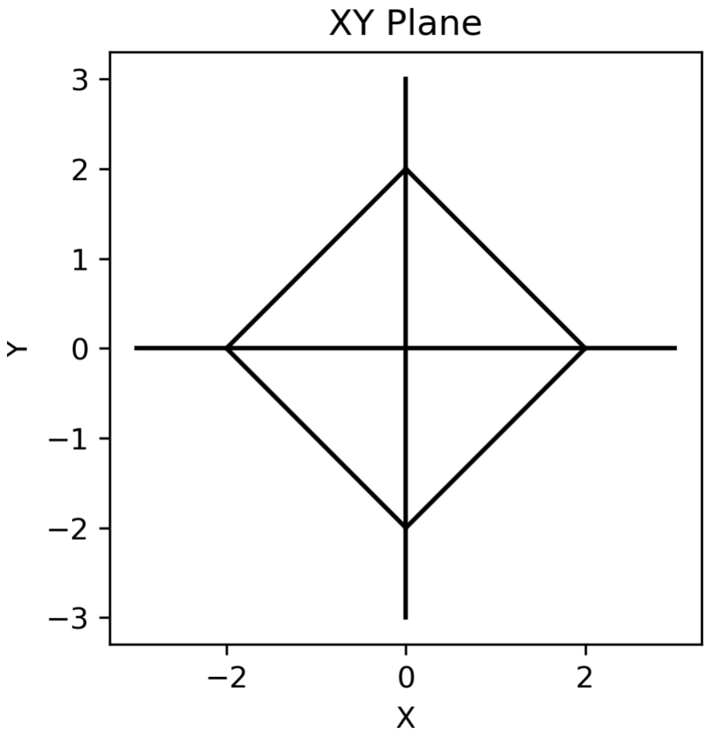
Option 3 — Non-Uniform Force Density

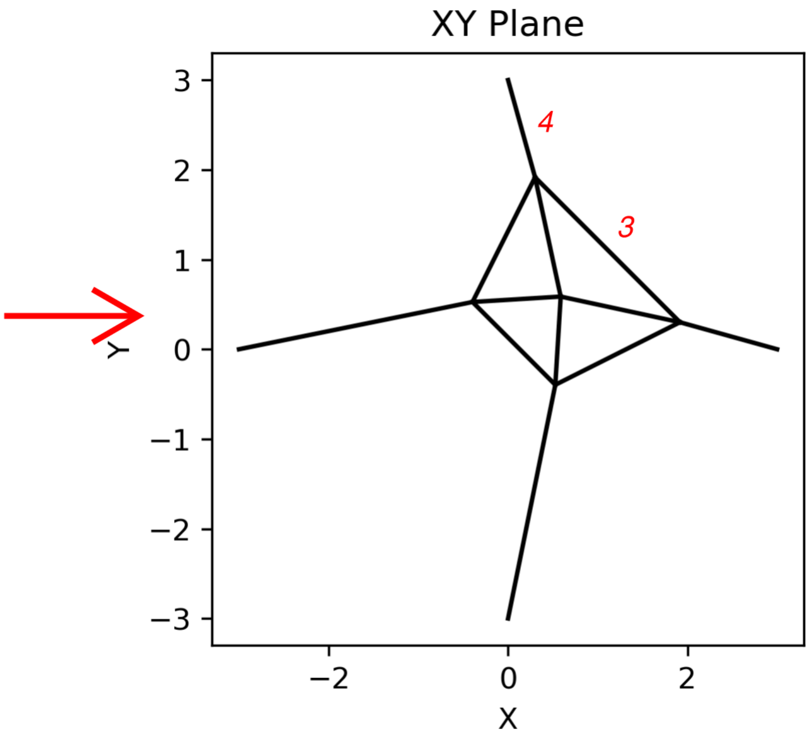
Assign different force densities to selected members:

$$q_i = \begin{cases} 4 & \text{for members 3 and 4} \\ 1 & \text{otherwise} \end{cases}$$

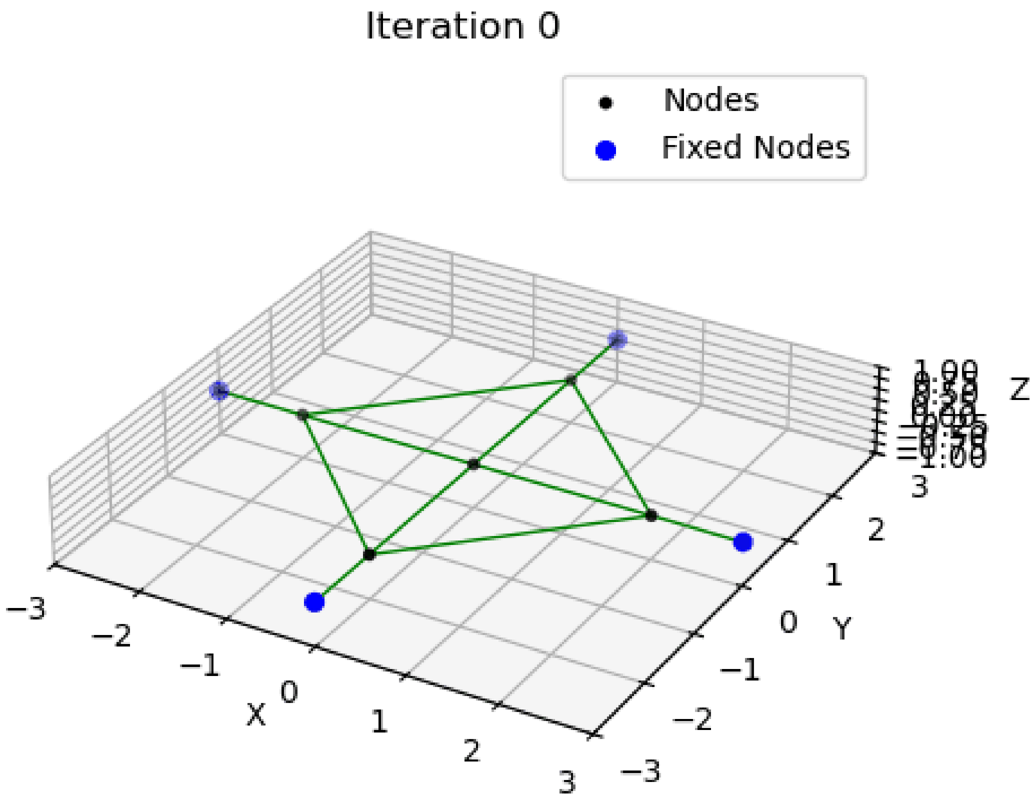
This changes the *relative* force densities.

Key idea:
The structure adapts its geometry based on relative q values — members with higher q tend to become shorter and stiffer in the final form.


XY Plane
X
Y
−2
0
2
3
1
−1
−3


XY Plane
4
3
X
Y
−2
0
2
3
1
−1
−3

Animation (only visible in Jupyter Notebook)

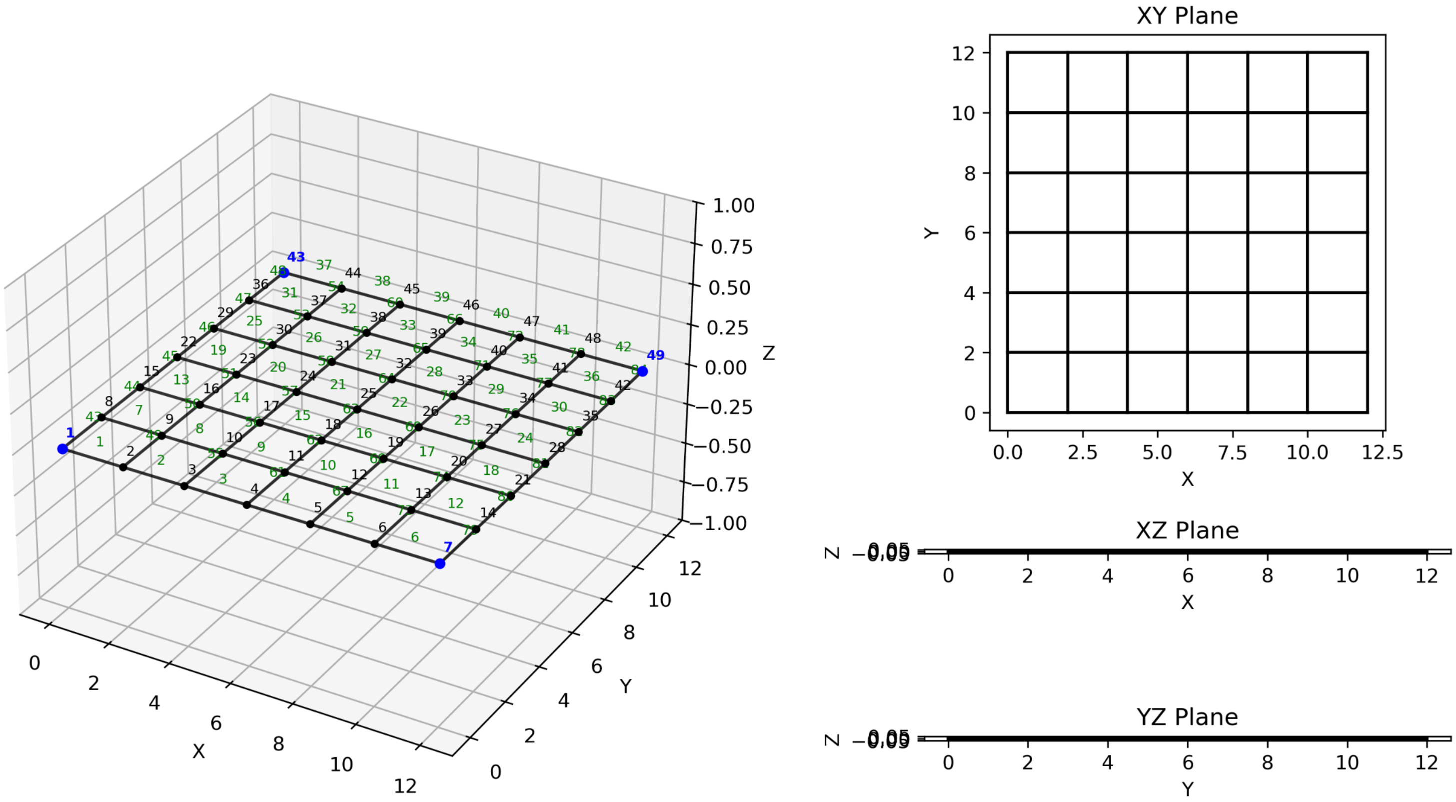


Option 4 — Other Force Density Pattern

Try assigning your own force density pattern. What sort of net structure do you get?

Part 2 - More Complex 2D Net

Description of Structure

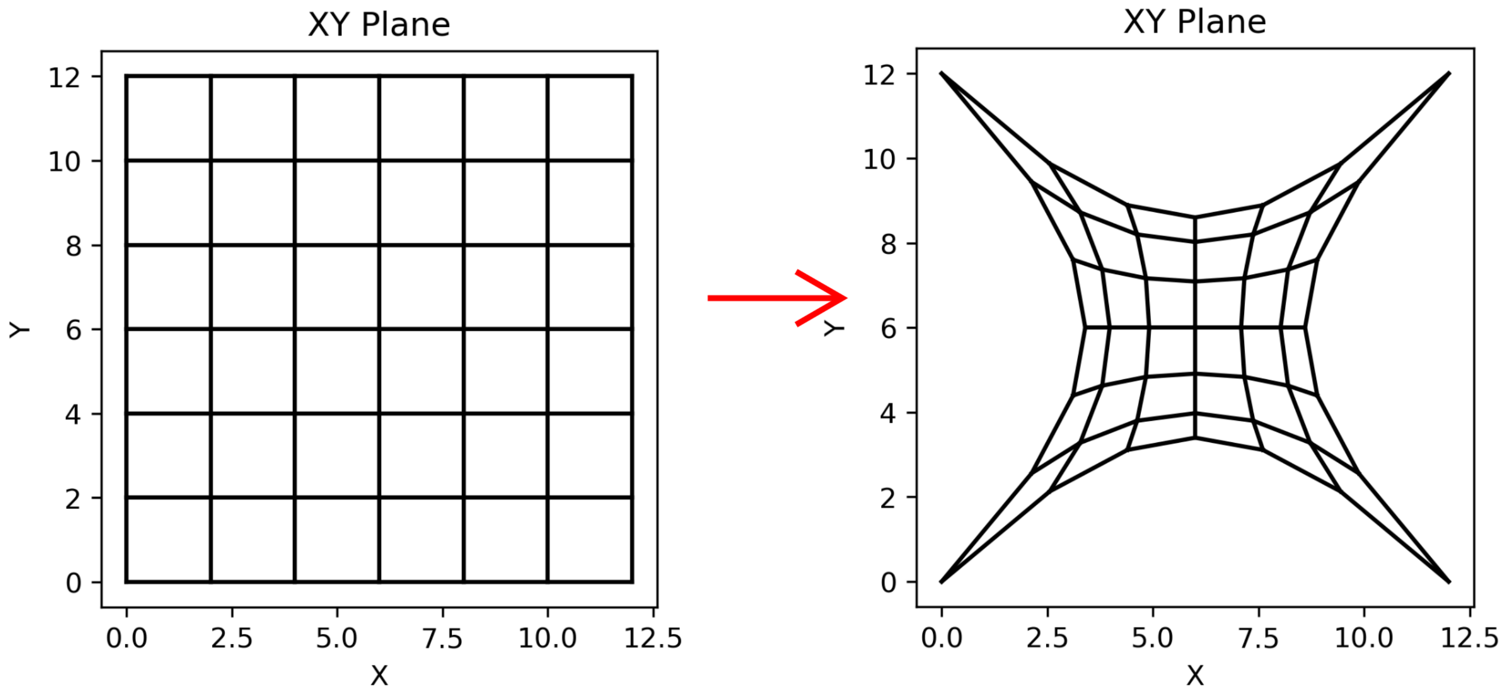


Option 1 — Uniform Force Density

Assume the same force density for every member: $q_i = 1$

So the force density vector becomes

$$\mathbf{q} = [1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1]^T$$


XY Plane
Y
12
10
8
6
4
2
0
0.0
2.5
5.0
7.5
10.0
12.5
X
XY Plane
Y
12
10
8
6
4
2
0
0.0
2.5
5.0
7.5
10.0
12.5
X

Option 2 — Edge-Dominated Force Density

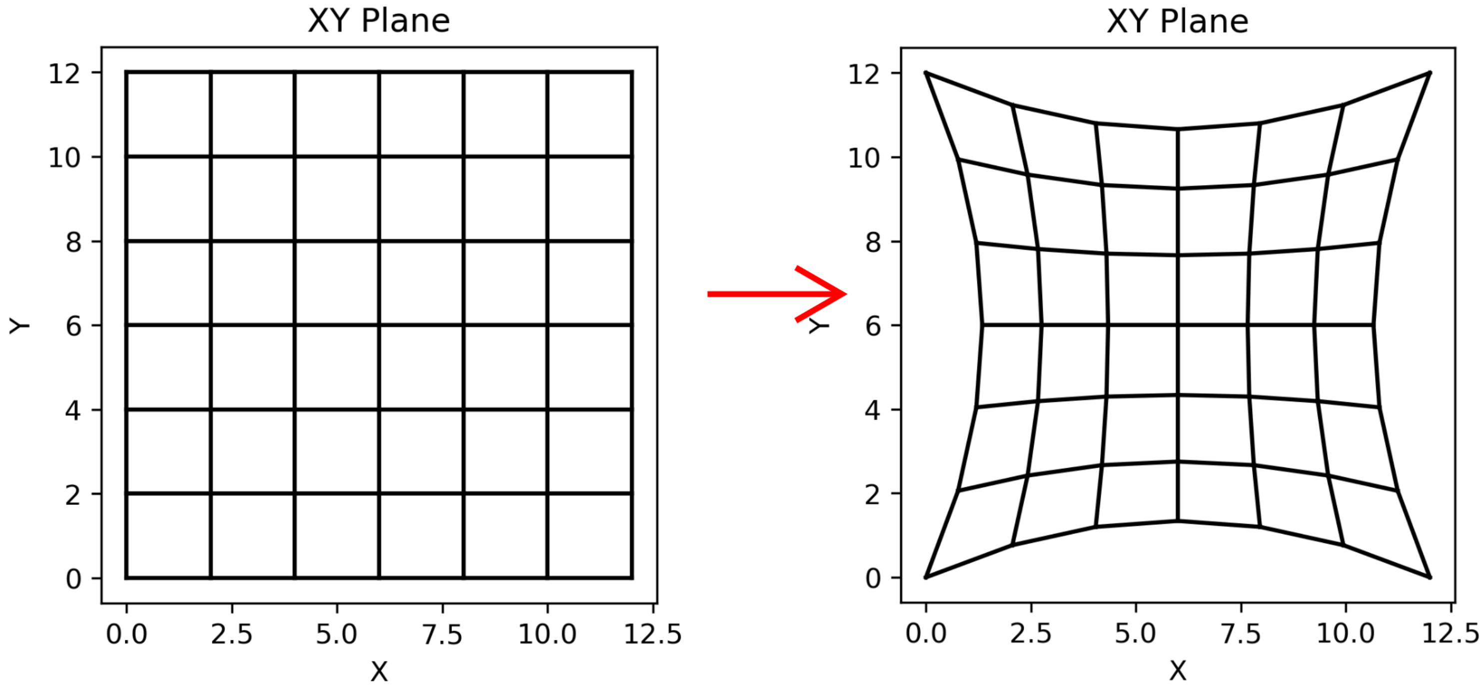
Assign larger force densities to the edge members than to the interior members:

$$q_{\text{edge}} = 5\, q_{\text{interior}}$$

So the ratio is

$$q_{\text{edge}} : q_{\text{interior}} = 5 : 1$$

Remember that only the relative values matter, what do you think making the edges stiffer than the interior will do?


XY Plane
Y
X
0
2
4
6
8
10
12
0.0
2.5
5.0
7.5
10.0
12.5
XY Plane
Y
X

Option 3 — Interior-Dominated Force Density

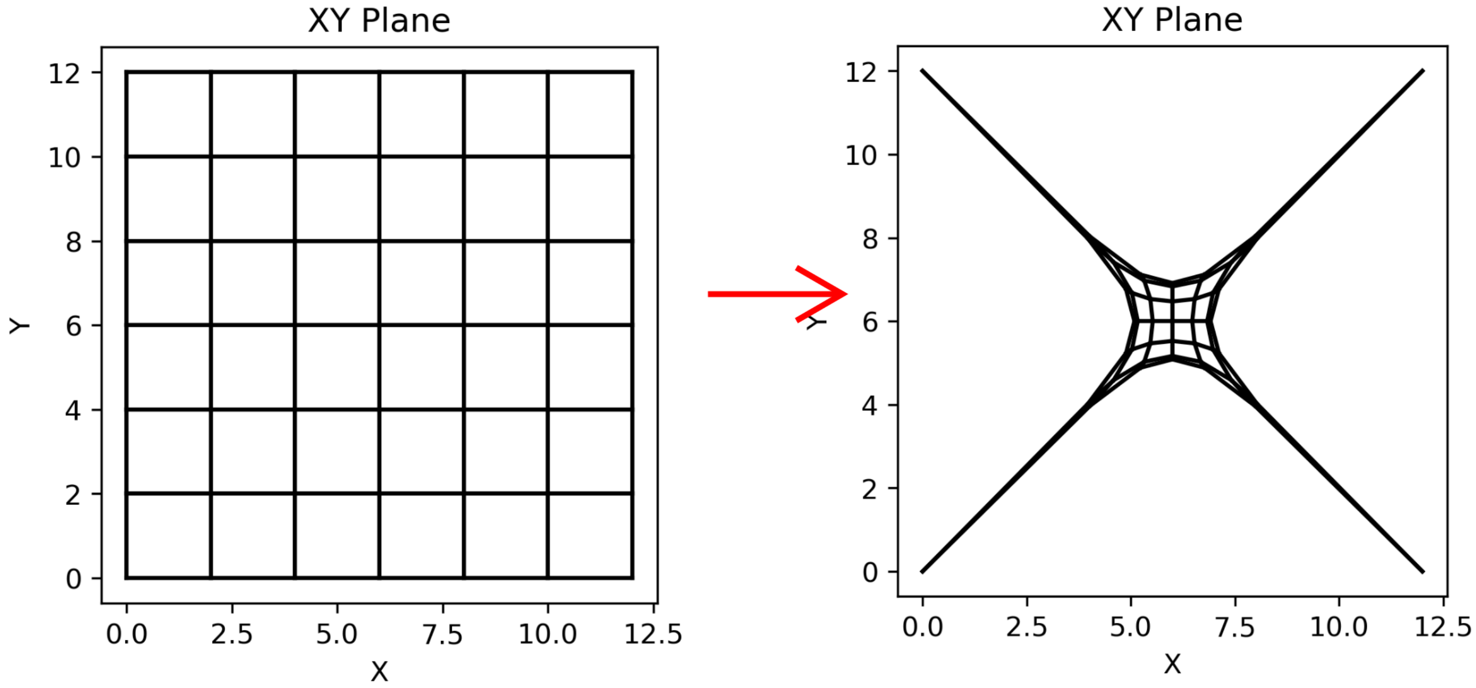
Assign larger force densities to the edge members than to the interior members:

$$q_{\text{interior}} = 5\, q_{\text{edge}}$$

So the ratio is

$$q_{\text{interior}} : q_{\text{edge}} = 5 : 1$$

Remember that only the relative values matter, what do you think making the interior stiffer than the edge will do?


XY Plane
12
10
8
6
4
2
0
Y
0.0
2.5
5.0
7.5
10.0
12.5
X
XY Plane
12
10
8
6
4
2
0
Y
0.0
2.5
5.0
7.5
10.0
12.5
X

Part 3 - Extend Force Density to 3D

Extension to 3D

For 3D form finding, we solve for the out-of-plane coordinates:

$$\mathbf{z} = \mathbf{D}^{-1}\left(\mathbf{p}_z - \mathbf{D}_f\mathbf{z}_f\right)$$

This is directly analogous to the x and y equations.

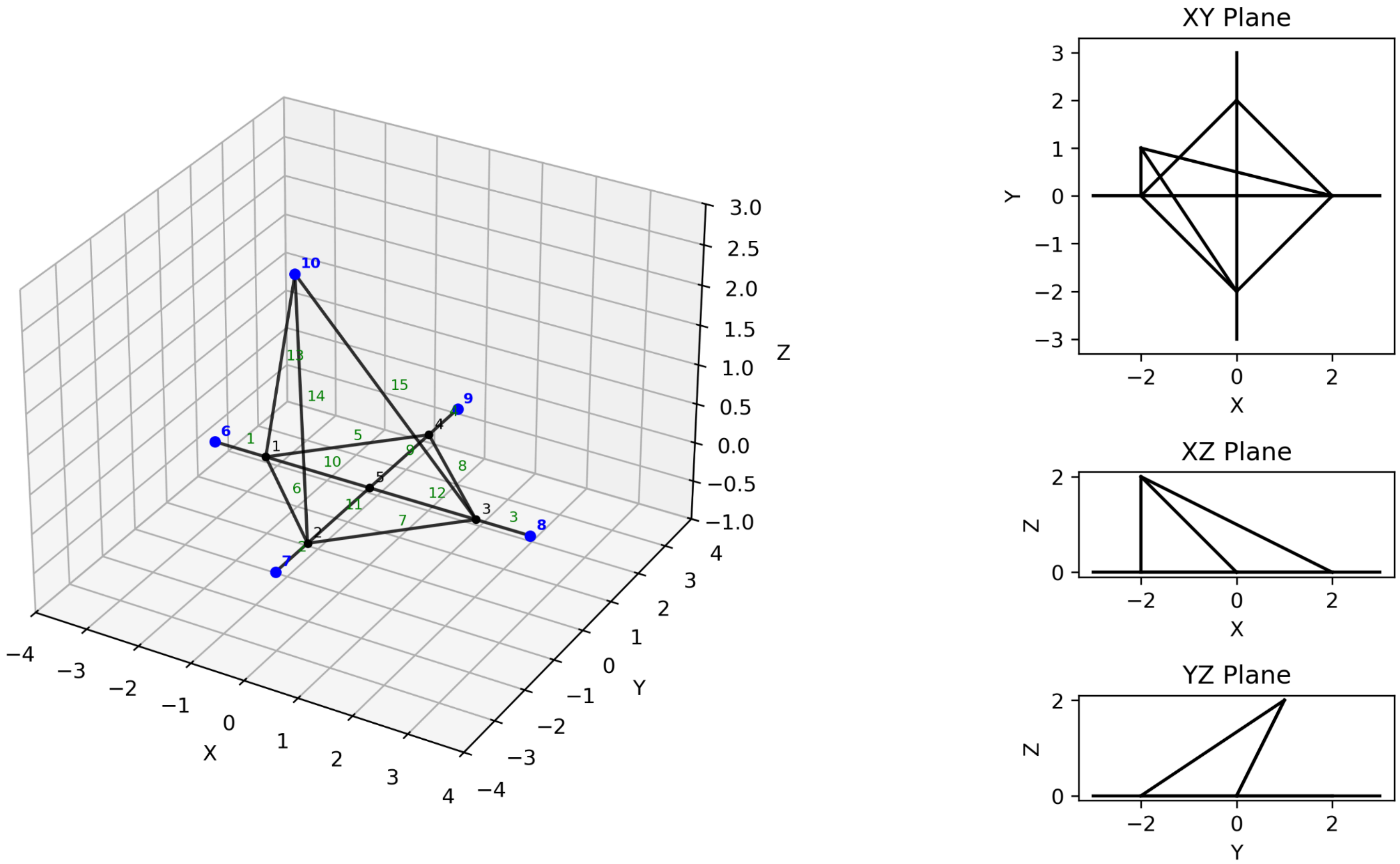
Key idea:
The same matrix $\mathbf{D}$ and $\mathbf{D_f}$ are used — we simply solve an additional system for the z-direction.

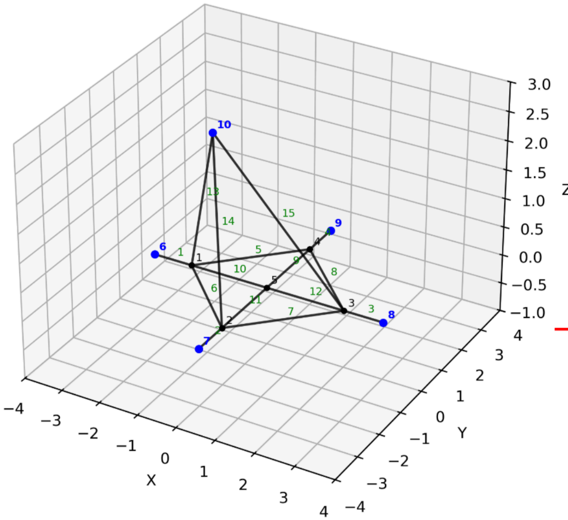
Option 1 — Schek Structure with an Extra Node

Add one extra node out of plane in the z-direction, connected to the structure by three additional members.

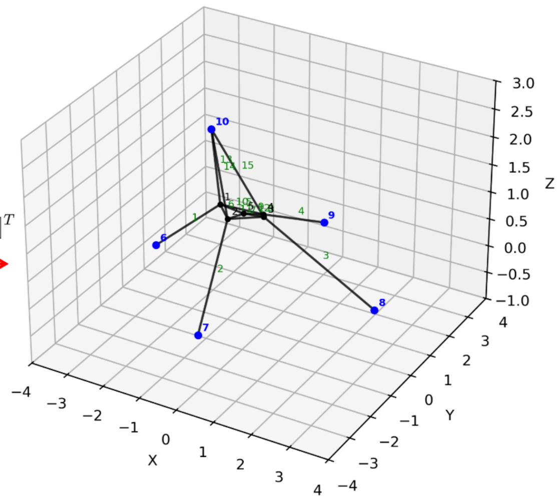
$$\mathbf{q} = [1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1]^T$$

$$\mathbf{q} = [3,\ 3,\ 3,\ 3,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1]^T$$

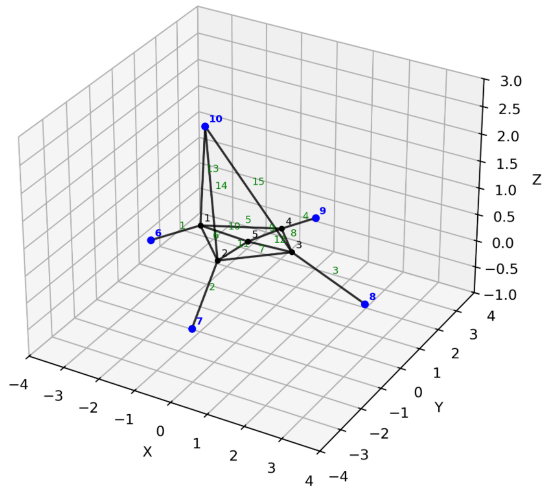

XY Plane
XZ Plane
YZ Plane
X
Y
Z



$$\mathbf{q} = [1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1]^T$$



$$\mathbf{q} = [3,\ 3,\ 3,\ 3,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1]^T$$



Option 2 — Veenendaal Validation Structure

A hyperbolic paraboloid cable-net with fixed boundaries is used as a validation example to compare all three form-finding methods:

- stiffness matrix method
- dynamic relaxation (with damping)
- force density method

The internal 3×3 mesh has 27 total DOFs.

All analyses start from an initially flat configuration.

Assumptions: $Q = 1$

Post-processing: Forces are normalized for comparison:

$$f_{\mathrm{norm}} = \frac{f}{F_{\mathrm{avg}}}$$

3. Minimal surface

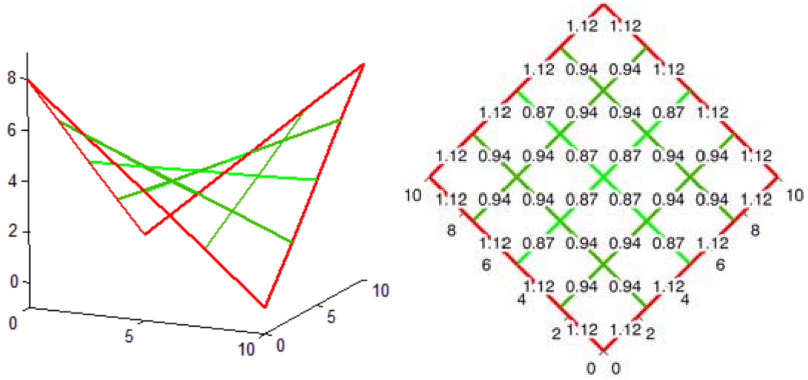
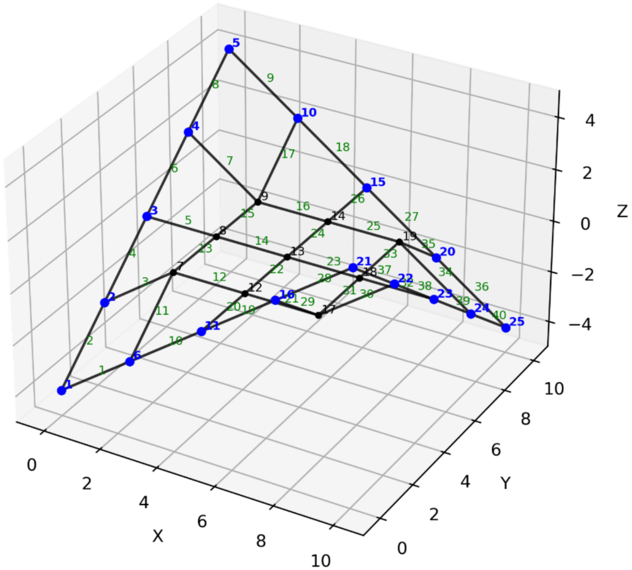


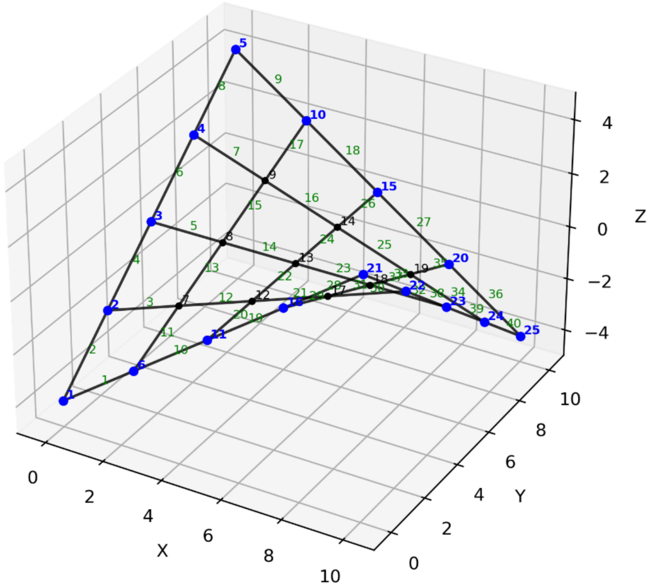
Fig. 1: Normalized forces $\mathbf{f}_{norm}$ *for* Q*=1, or* EA*=*F_o*=1*

As an example of the framework presented here, the stiffness matrix, dynamic relaxation with viscous and kinetic damping [9] and force density methods are compared by form finding a hyperbolic paraboloid cable-net (Figure 1) with fixed boundaries. Three mesh sizes are computed, with an increasing number of degrees of freedom (3 per node): n=3x3, 27 dof's (Figure 1), n=7x7, 147 dof's and n=15x15, 675 dof's. The form finding starts from an initially flat net (Table 2).

As a first trial, trivial input parameters are used: Q=1, EA=F_0=1 and all unstressed lengths L_0 are equal. The solution is considered to have converged when the squared sum of all residual forces changes by less than 1% between iterations, or when the residual forces are 0. The forces in the results are normalized to their average for comparison ($\mathbf{f}_{\text{norm}}=\mathbf{f}/F_{\text{avg}}$).



$Q = 1$



Option 3 — Pauletti Validation Structure

Validation example from:

Pauletti & Fernandes (2020) — *An outline of the natural force density method and its extension to quadrilateral elements*
(see L13 folder for the paper)

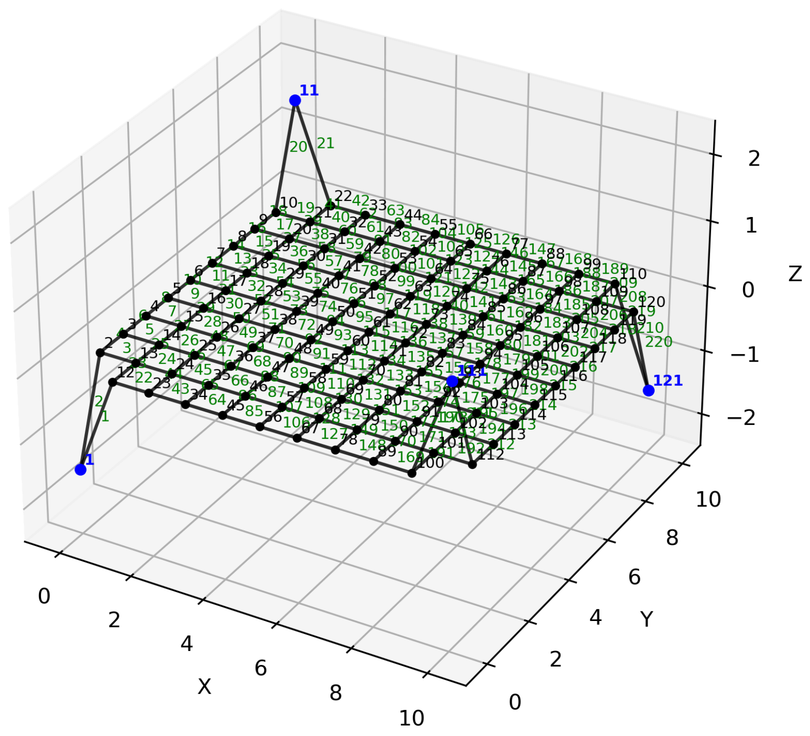
A dense cable net with pinned supports at opposite corner nodes.
The geometry is extended to 3D by offsetting the corners in the z-direction.

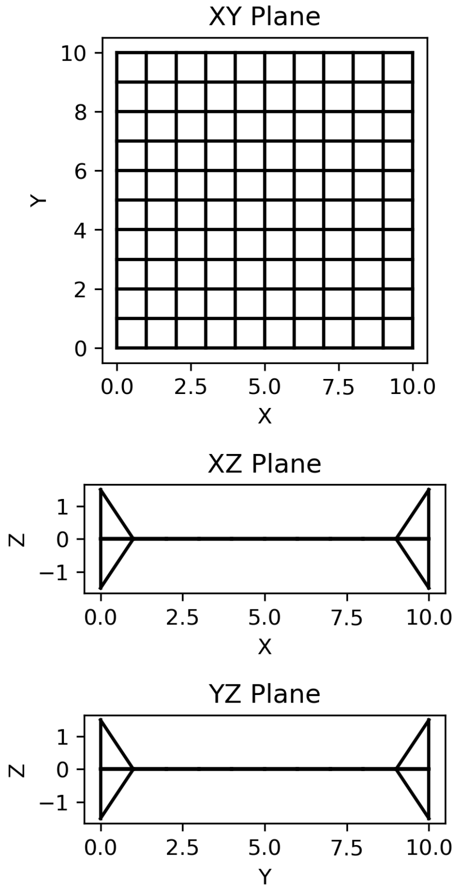
This setup is similar to the 2D "Complex Net", but introduces out-of-plane form finding.

Study parameter:

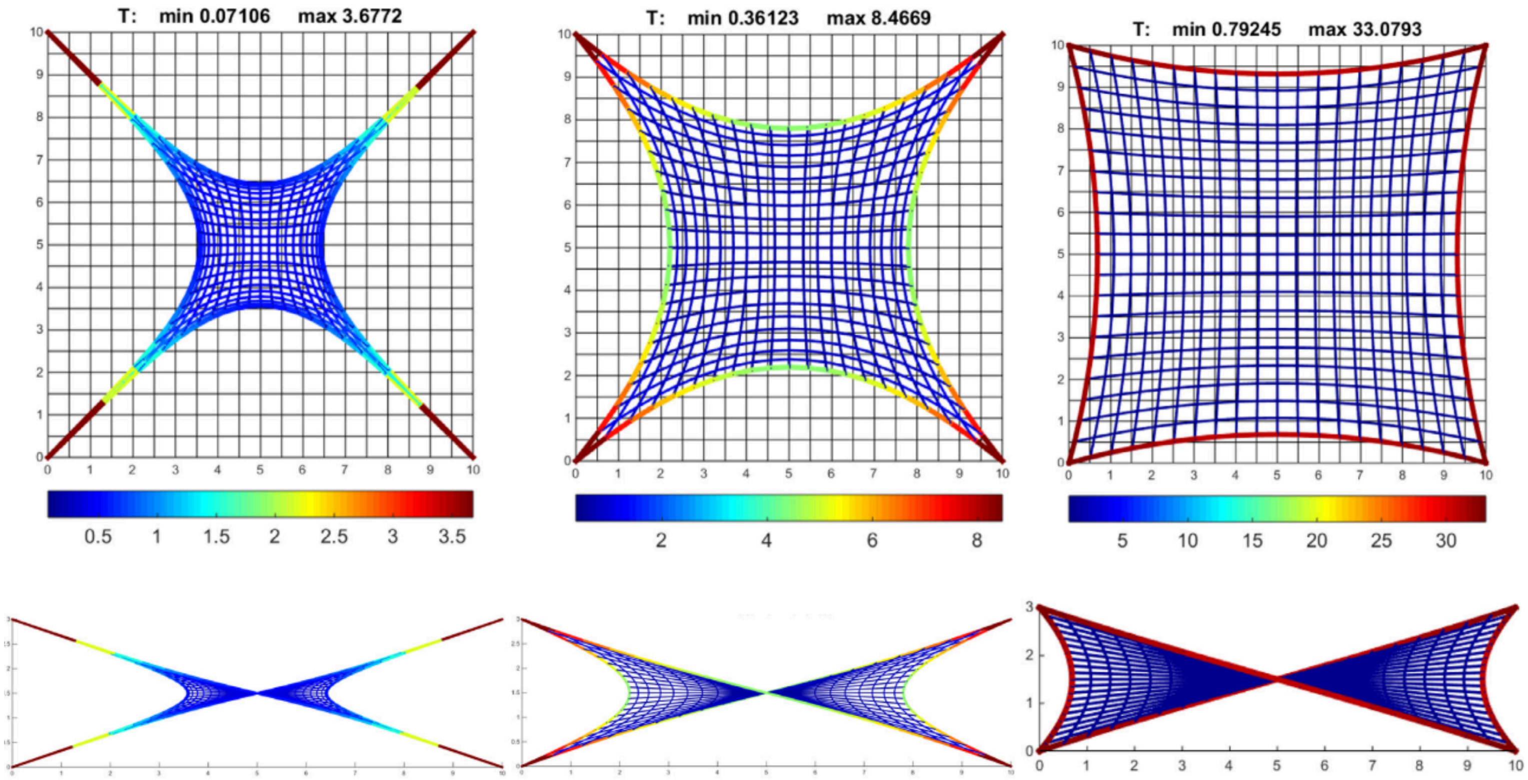
- Ratio of outer to inner force densities:

$$q_{\text{outer}} : q_{\text{inner}} = 1:1,\ 5:1,\ 30:1$$


X
Y
Z
0
2
4
6
8
10
−2
−1
1

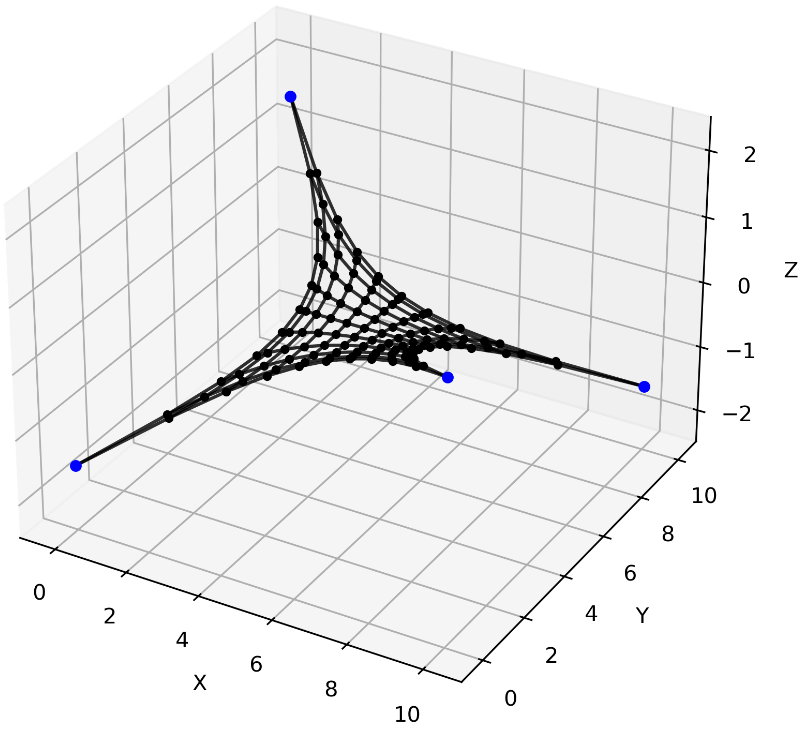

XY Plane
XZ Plane
YZ Plane
X
Y
Z
0.0
2.5
5.0
7.5
10.0

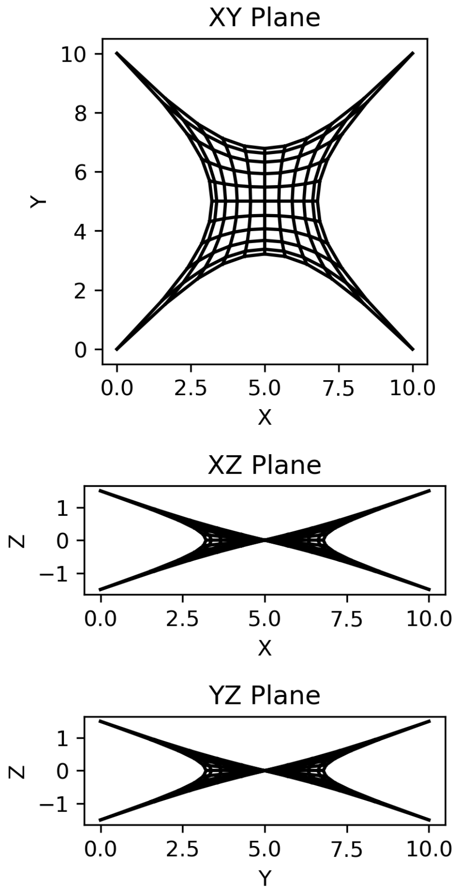
Results from Pauletti Paper



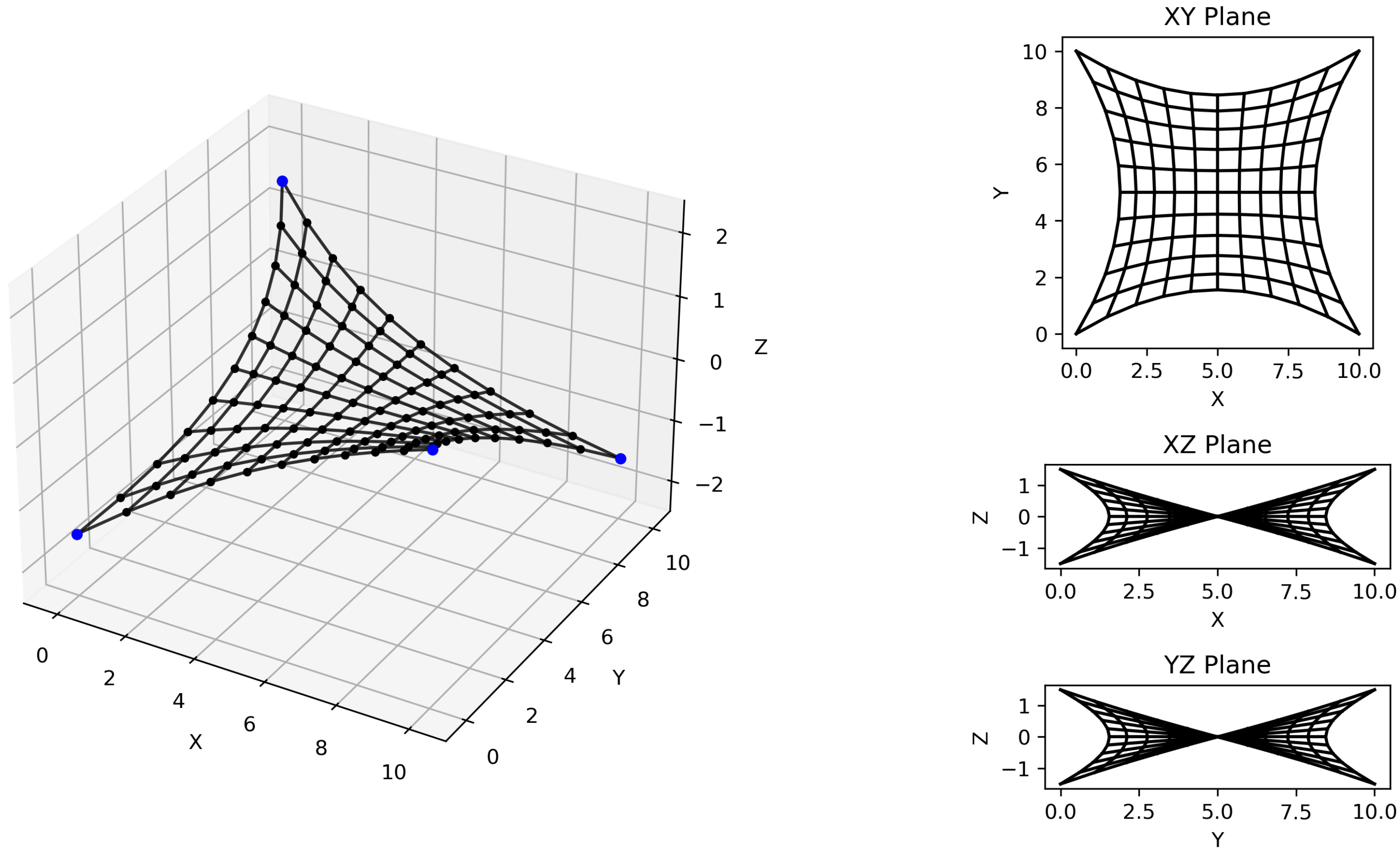
$N_b = 1; N_i = 1$ $N_b = 5; N_i = 1$ $N_b = 30; N_i = 1$

1:1

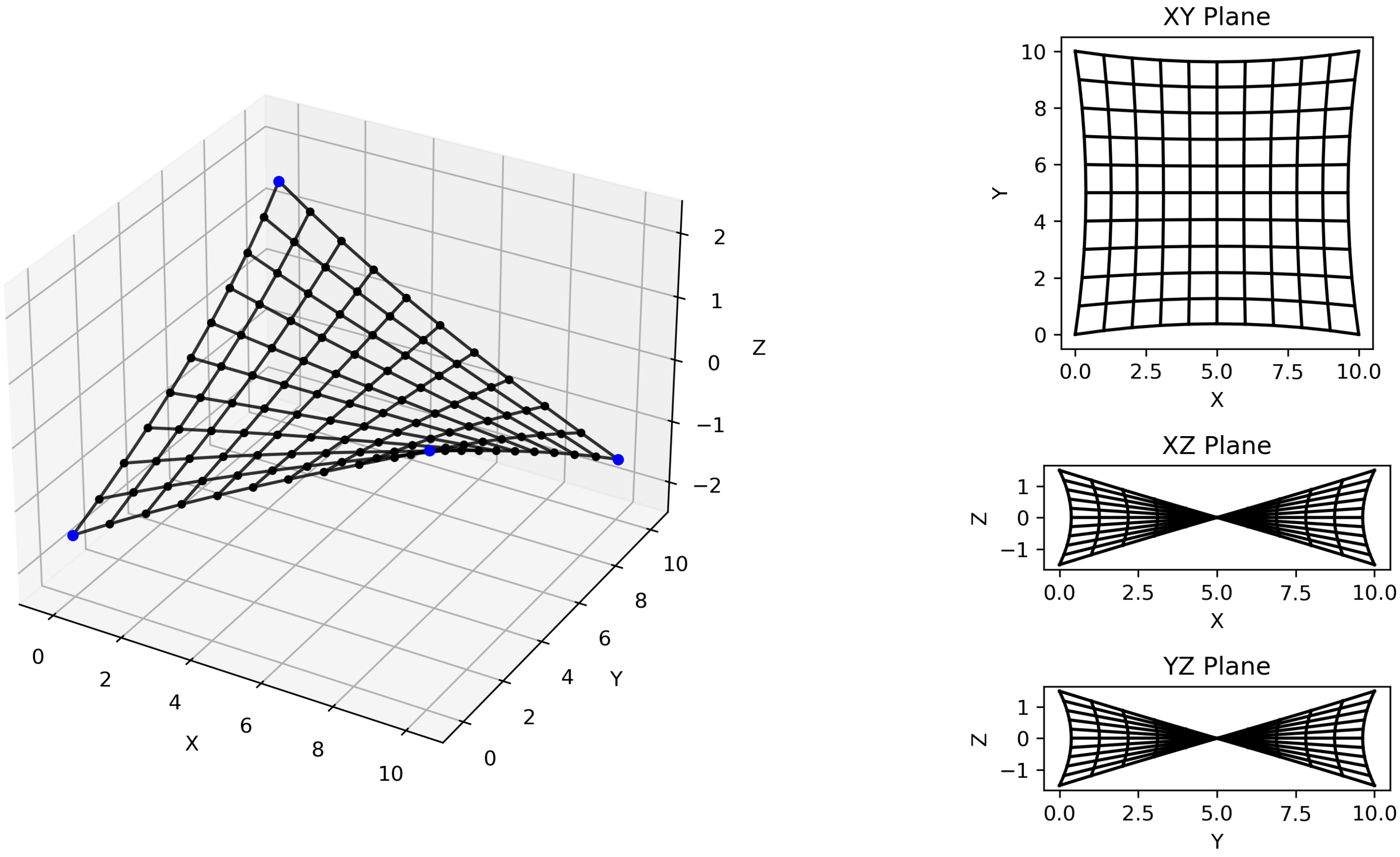




5:1



30:1



Part 4 - Adding Applied Loads

CEE6501 Point Load Example

So far, we have only considered form-finding problems in which the internal force state is prescribed through the force density values. You can think of this as a specified prestress or pretension in the network.

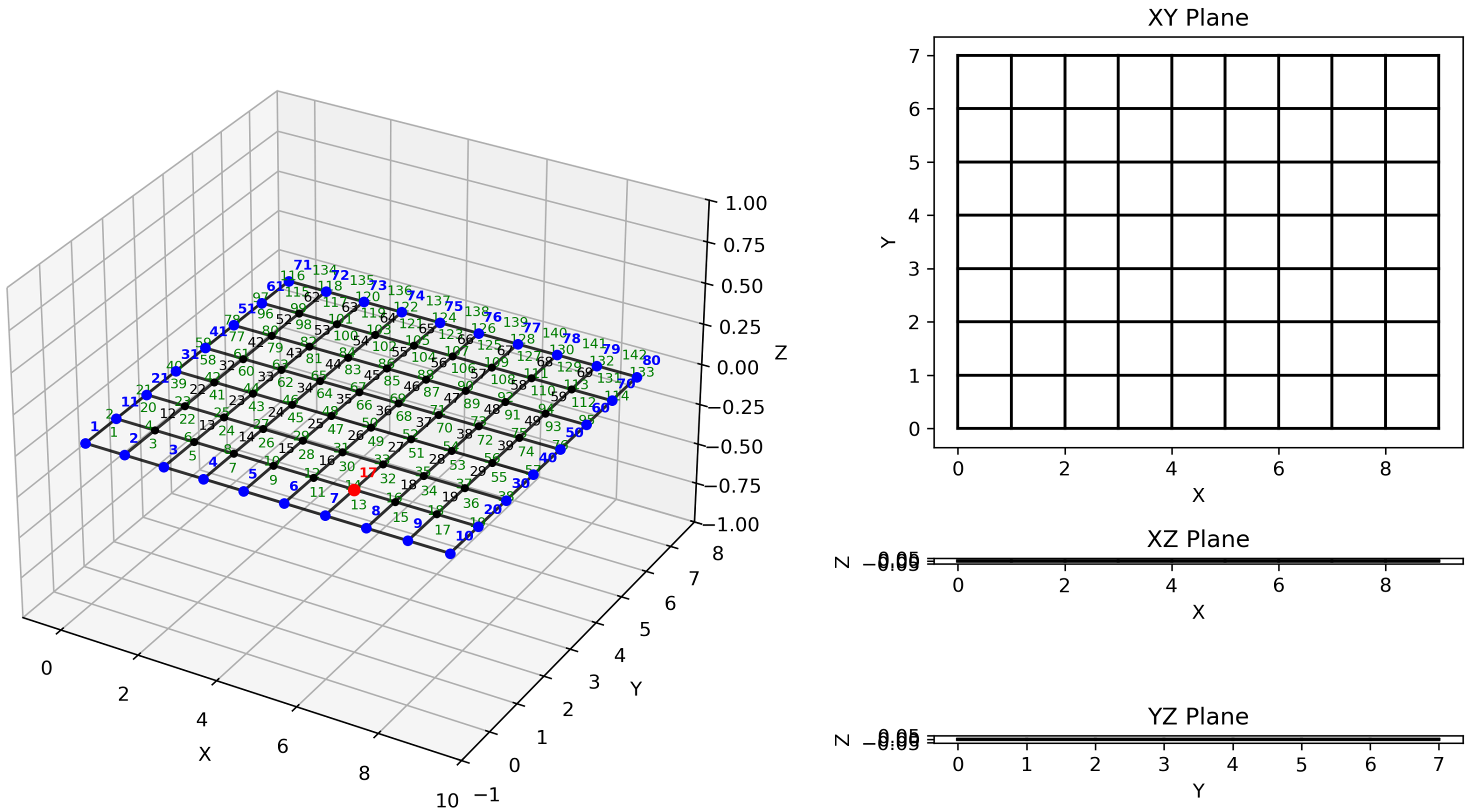
We can also include external applied loads, so the structure must now find a geometry that satisfies equilibrium under both:

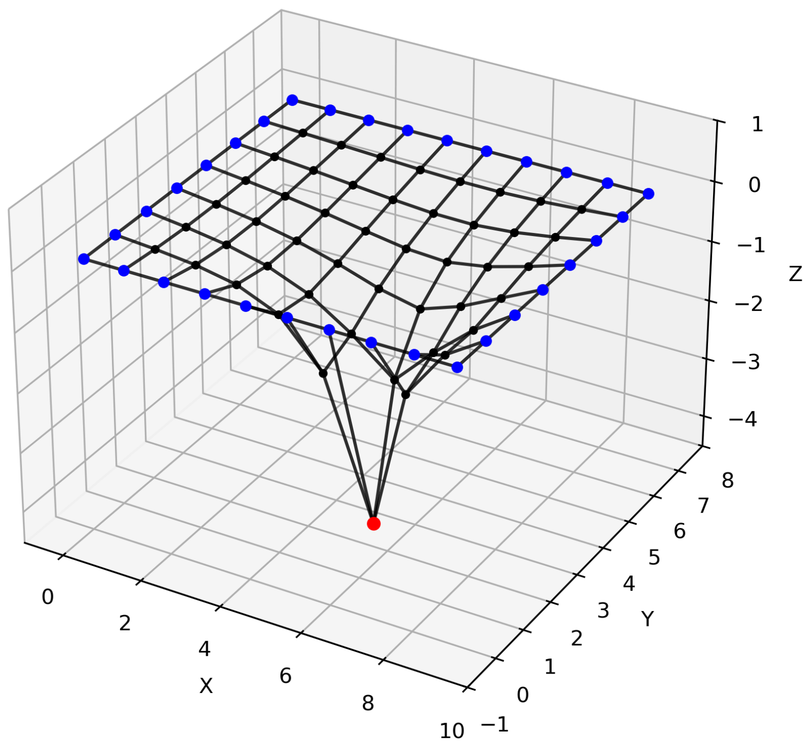
- the prescribed internal force density state, and
- the external nodal loads

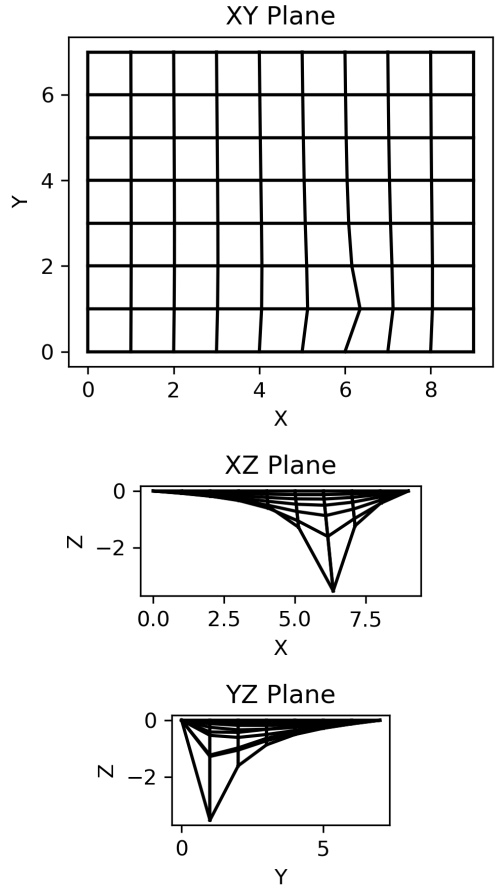
Question:
What shape does the net take under these combined conditions?

In this example, we prescribe a uniform force density, $q_i = 1$, in all members, and also apply downward point loads at selected nodes.

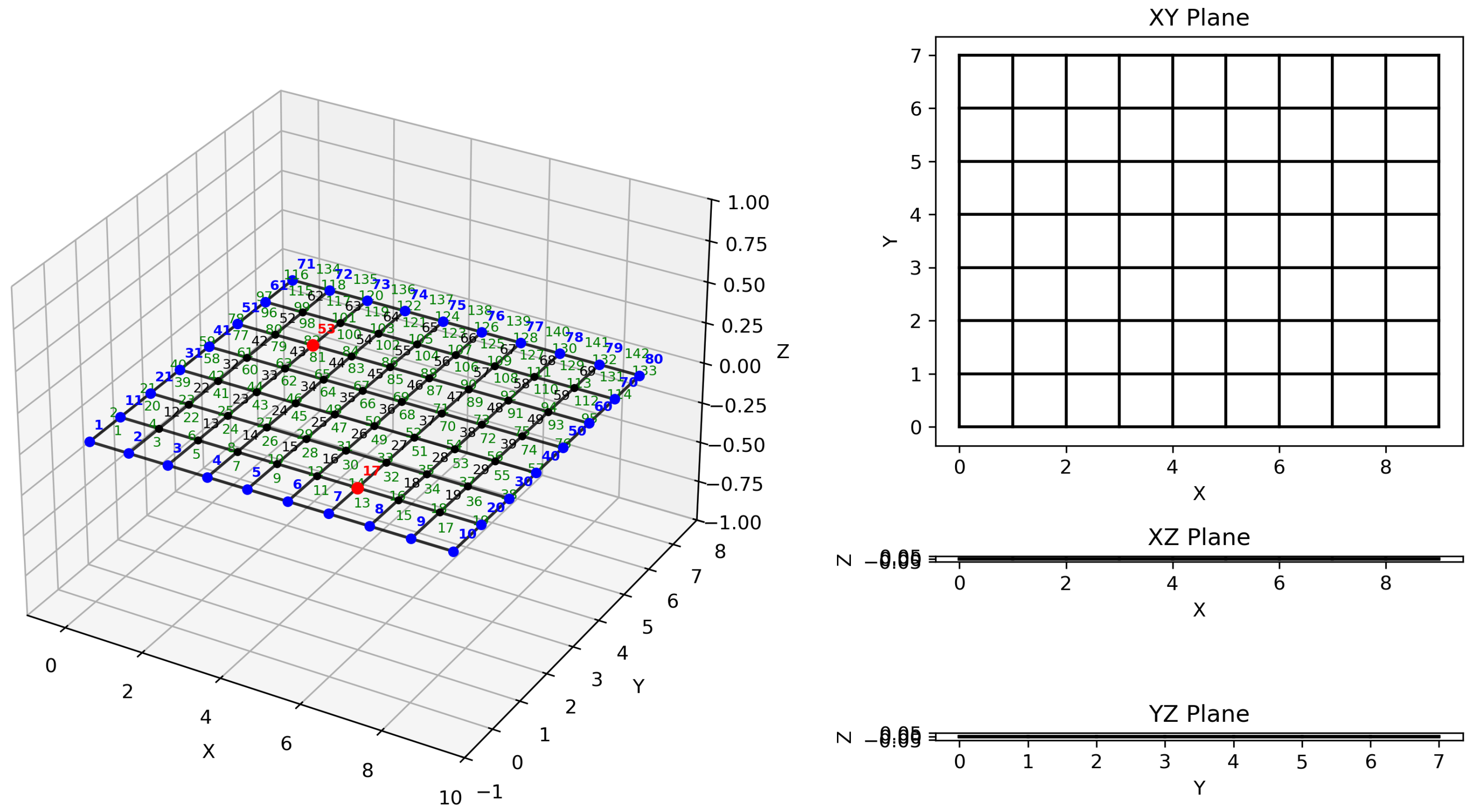
Single Point Load (2.0, 0.0, -10.0) at node 17

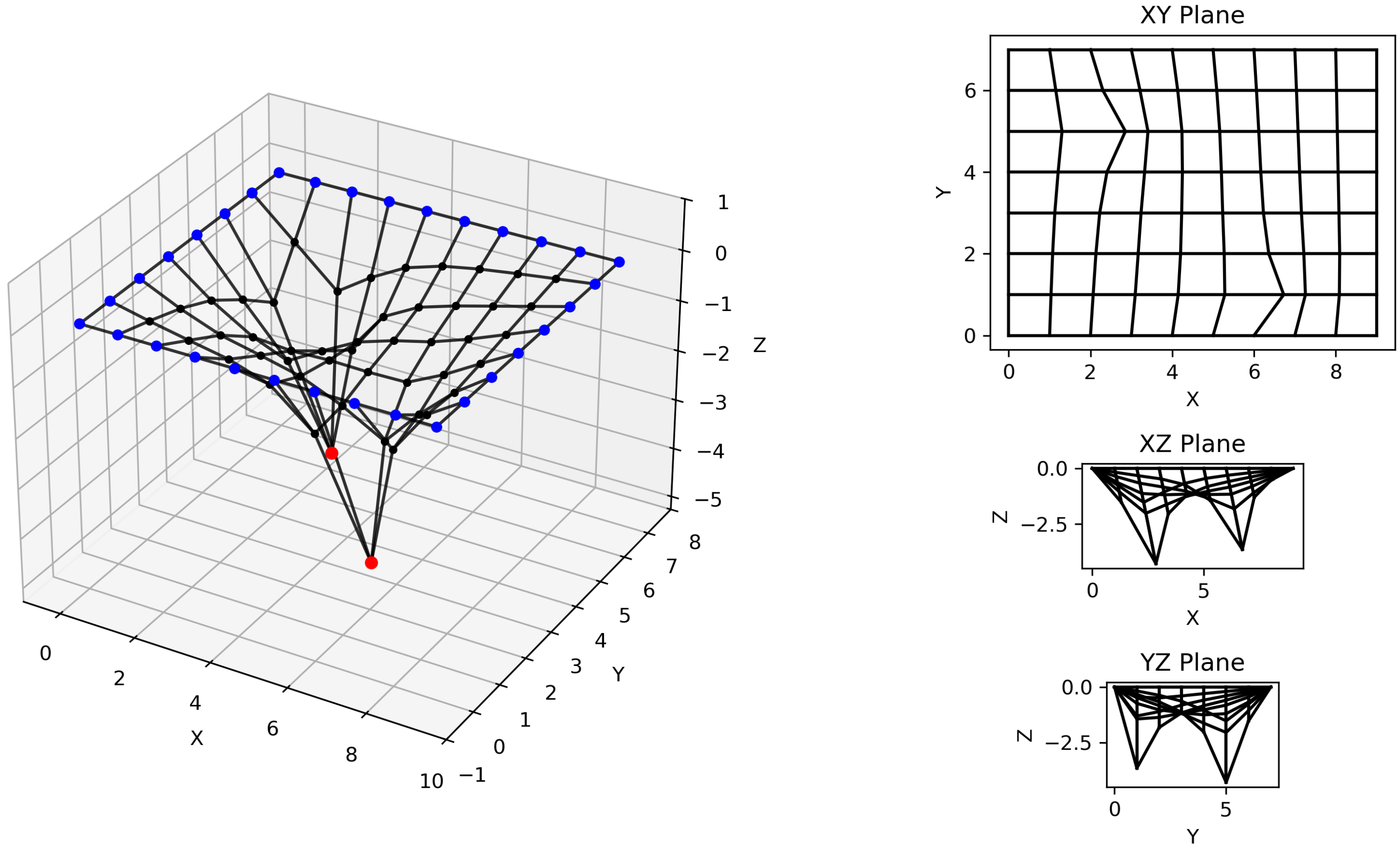



X
Y
Z
0
2
4
6
8
10
−1
1
3
5
7
−4
−3
−2


XY Plane
Y
X
XZ Plane
Z
X
0.0
2.5
5.0
7.5
YZ Plane
Z
Y

Two Point Loads (2.0, 0.0, -10.0) at node 17 and 53




XY Plane
XZ Plane
YZ Plane
X
Y
Z

Wrap-Up

In this lecture, we:

- placed the force density method within the broader form-finding framework discussed by Veenendaal and Block
- introduced Schek's force density method as a network-based form-finding approach
- defined the branch-node matrix and its partition into free-node and fixed-node parts
- derived the linear coordinate equations using force densities and branch equilibrium
- worked through a simple planar example and recovered the final geometry, branch lengths, and branch forces

Main takeaway: Once the connectivity, supports, loads, and force densities are prescribed, the force density method finds the equilibrium geometry through a linear solve.